Module 1

an OS is a interface between the user and the Computer

OS Security is critical to reduce an organisations attack surface

Powering on the computer starts the bootloader which starts the OS

User -> application -> OS -> hardware

The OS manages resource allocation

Module 2

Linux

- GUI or CLI
- multi-user
- Shell enviroment
- Package managers
- FHS (Filesystem hireachy system)
- Kernel -> processes and memory

Distributions

different versions, different tools and interfaces

Kali is often used for cybersecurity work.

shells include:

- bash
- ksh
- csh
- tcsh
- zsh

Module 3

This course was bash

FHS organises data

The root is the highest level directory

pwd - print working directory
ls - list directory
cd - change directory
cat - display file
head - display beginning

/etc - configs
/bin - binaries
/mnt - mount
/tmp - temp

grep - search for string
| - send output of command as input to command

find - search for criteria
- -name
- -iname
- -mtime

mkdir - make dir
rm - remove
touch - create file
cp - copy
mv - move

Perms

Linux files have 3 permissions:
- read
- write
- execute

users are part of groups

owner group other

e.g drwxrwx rwx

directory | rwx perms for user | for group | for other

ls -l displays permissions
ls -la display all files + perms

chmod changes permissions

chmod g+w, o-r file.txt

add write to group | remove read from other | file to apply to

users

root - elevated privileges
sudo - temporarily grant root

useradd
userdel
usermod

help

man - short for manual
↳ displays info about other commands

whatis - command description

explainshell

apropos - search man for a string

Module 4

Databases

Databases are large collections of data that can be accessed by multiple users at the same time.

Relational databases are databases made up of related tables

fields and records

Primary key - unique entry
Foreign key - primary key in another table

SQL

Structured query language lets us fetch data from the database

SELECT - which columns to return
FROM - which table to get them from
ORDER BY - sort
WHERE - filter
LIKE % - wildcard search
_ - wildcard for 1 char

Datatypes

- String
- numeric
- date and time

<> not equal

BETWEEN - date filter
AND
OR
NOT

table joins

table.field

INNER JOIN - rows that match on a field from order table

FROM ltable
INNER JOIN rtable ON ltable.pkey = rtable.fkey

LEFT JOIN - all of table 1 but only matching table2
RIGHT JOIN - all of table 2 but only table 1 matching
OUTER JOIN - all from Both

COUNT
AVG
SUM
} aggregate functions